

THETIS

N4EAC EDITION

RC1

RC1 MODIFICATION RECORD

FFT MAX for Legacy Meters
HF Band Panel Simplification

HERMES LITE 2

Expert Electronics

Power



Phones

MIC 1

MIC 2

Pwr

SWR

A

B

A MODIFICATION GUIDE BY

N4EAC

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MAY 2026 • THETIS N4EAC EDITION RC1

RC1 Modification Record

Thetis N4EAC Edition RC1 - FFT Max Bin legacy meter support and HL2-focused HF band panel simplification

Document field	Value
Edition	Thetis N4EAC Edition
Release	RC1
Author / Creator	Eduardo de Carvalho (N4EAC)
Base project	OpenHPSDR Thetis v2.10.3.15 source tree
Primary target radio	Hermes Lite 2 (HL2)
License context	GNU General Public License Version 2 (GPLv2)
Document purpose	Provide a reproducible record of RC1 modifications.

Important accuracy note: Line numbers in this document are based on the RC1 working tree captured during the development session. Upstream Thetis line numbers may differ. When reproducing the changes, use the file names, method names, and search anchors shown in each section as the primary guide.

Scope: This document records the modifications made to add FFT Max Bin as a selectable source for the legacy RX1/RX2 meters, update the meter display paths, simplify the HF band panel for Hermes Lite 2 operation, and apply RC1 branding.

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1. Overview and Design Goals

RC1 had two functional goals and one release-management goal. The functional goals were to expose the existing FFT Max Bin detector in the legacy RX1 and RX2 meter user interface, and to simplify the HF band panel for a Hermes Lite 2 workflow. The release-management goal was to create a traceable custom edition identified as Thetis N4EAC Edition RC1.

- Add "FFT Max" to the legacy RX1 meter drop-down.
- Add "FFT Max" to the legacy RX2 meter drop-down.
- Use Thetis existing FFT Max detector path rather than inventing a new DSP measurement.
- Make the analog scale and digital dBm/SMeter text paths recognize the new meter mode.
- Hide band keys not used in the HL2-focused workflow: 6m, 2m, and VHF+.
- Move WWV and SWL so the HF band panel looks intentional and compact.
- Add RC1 branding to the window title and About dialog.

The important architectural discovery in RC1 was that FFT Max Bin is already computed by Thetis in the modern meter infrastructure via `Reading.SIGNAL_MAX_BIN` and `WDSP.GetDetectMaxBin(...)`. The legacy meter path did not expose it. RC1 therefore connects existing functionality to the legacy meter UI instead of implementing a new detector.

2. Prerequisites and Safety Workflow

Before applying RC1 changes to a clean Thetis source tree, prepare the following environment.

- Windows 11 or Windows 10 development machine.
- Microsoft Visual Studio 2022 with .NET desktop development and Desktop development with C++ workloads.
- The original OpenHPSDR Thetis source tree.
- A clean backup of the source folder before editing.
- Git or GitHub Desktop for source control.

Recommended safety workflow:

- Build the unmodified source first.
- Retarget native C++ projects to the Visual Studio 2022 v143 toolset if they still reference v145.
- Make one logical edit at a time.
- Build after each edit group.
- Run Thetis and test the edited feature before moving on.
- Commit a known-good checkpoint before starting the next RC.

Build configuration used during RC1: Release x64. Native helper projects such as `portaudio`, `wdsp`, `cmASIO`, and `ChannelMaster` were retargeted from v145 to Visual Studio 2022 v143 when needed.

3. Files Modified in RC1

File	Purpose
Project Files/Source/Console/enums.cs	Adds the new <code>MeterRXMode.SIGNAL_MAX_BIN</code> enum value.
Project Files/Source/Console/console.cs	Adds FFT Max to RX1/RX2 combo boxes, mapping handlers, meter update paths, analog/digital display.
Project Files/Source/Console/frmAbout.cs	Adds RC1 edition information and custom enhancement list to the About dialog.
README.md	Documents the custom edition and RC1 enhancements.
CHANGELOG.md	Records RC1 release notes.

The core functional changes are in `enums.cs` and `console.cs`. The About dialog and README changes are release-polish items, but they are important for identifying this as a modified GPLv2 build.

4. FFT Max Bin Implementation

4.1 Add the RX Meter Enum Value

File: Project Files/Source/Console/enums.cs

Approximate RC1 line: 164-175

Search anchor: public enum MeterRXMode

The legacy RX meter mode is driven by the MeterRXMode enum. RC1 adds SIGNAL_MAX_BIN after OFF and before LAST. This keeps the original existing modes in their original order.

```
public enum MeterRXMode
{
    FIRST = -1,
    SIGNAL_STRENGTH,
    SIGNAL_AVERAGE,
    ADC_L,
    ADC_R,
    ADC2_L,
    ADC2_R,
    OFF,
    SIGNAL_MAX_BIN,
    LAST,
}
```

Why: The combo boxes and meter update logic all exchange values using MeterRXMode. Without this enum value, "FFT Max" cannot be represented in the legacy meter pipeline.

4.2 Add FFT Max to RX1 and RX2 Meter Drop-Downs

File: console.cs

Approximate RC1 lines: 5649-5667

Search anchor: private void InitMultiMeterModes()

RC1 adds "FFT Max" to both RX1 and RX2 meter mode lists. It is placed after "Sig Avg" so the three signal-related modes appear together.

```
private void InitMultiMeterModes()
{
    comboMeterRXMode.Items.Add("Signal");
    comboMeterRXMode.Items.Add("Sig Avg");
    comboMeterRXMode.Items.Add("FFT Max");
    comboMeterRXMode.Items.Add("ADC L");
    comboMeterRXMode.Items.Add("ADC R");
    comboMeterRXMode.Items.Add("ADC2 L");
    comboMeterRXMode.Items.Add("ADC2 R");
    comboMeterRXMode.Items.Add("Off");

    comboRX2MeterMode.Items.Add("Signal");
    comboRX2MeterMode.Items.Add("Sig Avg");
    comboRX2MeterMode.Items.Add("FFT Max");
    comboRX2MeterMode.Items.Add("ADC L");
    comboRX2MeterMode.Items.Add("ADC R");
    comboRX2MeterMode.Items.Add("ADC2 L");
    comboRX2MeterMode.Items.Add("ADC2 R");
    comboRX2MeterMode.Items.Add("Off");
}
```

Why: This exposes the new mode in the UI. It does not yet make the meter work; later sections wire the selection to actual data and rendering paths.

5. Selection Handling and Text Mapping

5.1 RX1 Property Setter Text Mapping

File: console.cs

Approximate RC1 lines: 18124-18136

Search anchor: public MeterRXMode CurrentMeterRXMode

The property setter translates enum values into the visible combo-box text. Add a complete case for FFT Max.

```
case MeterRXMode.SIGNAL_AVERAGE:
    text = "Sig Avg";
    break;
case MeterRXMode.SIGNAL_MAX_BIN:
    text = "FFT Max";
    break;
case MeterRXMode.ADC_L:
    text = "ADC L";
    break;
```

5.2 RX2 Property Setter Text Mapping

File: console.cs

Approximate RC1 lines: 18180-18193

Search anchor: public MeterRXMode RX2MeterMode

Make the equivalent text mapping change for RX2.

```
case MeterRXMode.SIGNAL_AVERAGE:
    text = "Sig Avg";
    break;
case MeterRXMode.SIGNAL_MAX_BIN:
    text = "FFT Max";
    break;
case MeterRXMode.ADC_L:
    text = "ADC L";
    break;
```

6. Combo-Box Selection Handlers

6.1 RX1 comboMeterRXMode_SelectedIndexChanged

File: console.cs

Approximate RC1 lines: 30550-30561

Search anchor: switch (comboMeterRXMode.Text)

When the user selects "FFT Max" from the RX1 meter drop-down, map that string to the new enum value.

```
case "Signal":
    mode = MeterRXMode.SIGNAL_STRENGTH;
    break;
case "Sig Avg":
    mode = MeterRXMode.SIGNAL_AVERAGE;
    break;
case "FFT Max":
    mode = MeterRXMode.SIGNAL_MAX_BIN;
    break;
case "ADC L":
    mode = MeterRXMode.ADC_L;
    break;
```

6.2 RX2 comboRX2MeterMode_SelectedIndexChanged

File: console.cs

Approximate RC1 lines: 39478-39488

Search anchor: switch (comboRX2MeterMode.Text)

Repeat the same string-to-enum mapping for RX2.

```
case "Signal":
    mode = MeterRXMode.SIGNAL_STRENGTH;
    break;
case "Sig Avg":
    mode = MeterRXMode.SIGNAL_AVERAGE;
    break;
case "FFT Max":
    mode = MeterRXMode.SIGNAL_MAX_BIN;
    break;
case "ADC L":
    mode = MeterRXMode.ADC_L;
    break;
```

7. Meter Data Acquisition

RC1 originally attempted to use a non-existent `WDSP.MeterType.SIGNAL_MAX_BIN`. The build correctly failed, showing that this Thetis branch does not expose FFT Max Bin through `WDSP.CalculateRXMeter(...)`. The correct path is to enable the existing display max-bin detector and then call `WDSP.GetDetectMaxBin(...)`.

7.1 RX1 UpdateMultimeter()

File: console.cs

Approximate RC1 lines: 24770-24785

Search anchor: private async void UpdateMultimeter()

Add the following case after `SIGNAL_AVERAGE` and before the ADC cases.

```
case MeterRXMode.SIGNAL_MAX_BIN:
    if (!_display_max_bin_enabled[0]) setupDisplayMaxBinDetect(1, false, true);
    num = (float)WDSP.GetDetectMaxBin(0);
    if (num <= -400.0f) num = -200.0f;
    num += RXOffset(1);
    new_meter_data = num;
    break;
```

Important: RX1 uses detector index 0 with `WDSP.GetDetectMaxBin(0)`. The call to `setupDisplayMaxBinDetect(1, false, true)` ensures the RX1 max-bin detector is enabled before reading it.

7.2 RX2 UpdateRX2Multimeter()

File: console.cs

Approximate RC1 lines: 24935-24951

Search anchor: UpdateRX2Multimeter or switch (rx2_meter_mode)

Add the RX2 equivalent case. The crucial discovery was that RX2 display setup uses receiver number 2, but the detector index used by WDSP.GetDetectMaxBin is 1.

```
case MeterRXMode.SIGNAL_MAX_BIN:
    if (!_display_max_bin_enabled[1]) setupDisplayMaxBinDetect(2, false, true);
    num = (float)WDSP.GetDetectMaxBin(1);
    if (num <= -400.0f) num = -200.0f;
    num += RXOffset(2);
    rx2_meter_new_data = num;
    break;
```

Why index 1: Testing showed that WDSP.GetDetectMaxBin(2) returned no useful RX2 data, leaving RX2 at -200 dBm. Changing it to WDSP.GetDetectMaxBin(1) made RX2 FFT Max track correctly.

8. Drawing and Digital Readout Paths

8.1 Analog Scale Drawing

File: console.cs

Approximate RC1 lines: 22507-22520

Search anchor: if (!_mox || rx == 2) and case MeterRXMode.SIGNAL_AVERAGE:

The analog meter scale was blank until the drawing switch was told to treat FFT Max as a signal-style meter. Add only the case label so it falls through into the same drawing block as Signal and Sig Avg.

```
switch (rxMode)
{
    case MeterRXMode.SIGNAL_STRENGTH:
    case MeterRXMode.SIGNAL_AVERAGE:
    case MeterRXMode.SIGNAL_MAX_BIN:
        if (bDrawMarkers)
        {
            g.FillRectangle(low_brush, 0, H - 4, (int)(W * 0.5), 2);
            g.FillRectangle(high_brush, (int)(W * 0.5), H - 4, (int)(W * 0.5) - 4, 2);
        }
        double spacing = (W * 0.5 - 2.0) / 5.0;
        ...
}
```

8.2 RX1 Digital Text Path

File: console.cs

Approximate RC1 lines: 23906-23915

Search anchor: switch (current_meter_rx_mode)

Add FFT Max as a fall-through label so the digital text formatting path treats it as a signal reading.

```
switch (current_meter_rx_mode)
{
    case MeterRXMode.SIGNAL_STRENGTH:
    case MeterRXMode.SIGNAL_AVERAGE:
    case MeterRXMode.SIGNAL_MAX_BIN:
        switch (m_eMeasureMode)
        {
            case MultiMeterMeasureMode.SMeter:
                output = Common.SMeterFromDBM(num, VFOAFreq >= S9Frequency);
                break;
            case MultiMeterMeasureMode.DBM:
                output = num.ToString(format) + " dBm";
                break;
        }
        break;
}
```


8.3 RX2 Digital Text Path

File: console.cs

Approximate RC1 lines: 24142-24155

Search anchor: switch (rx2_meter_mode)

```
switch (rx2_meter_mode)
{
    case MeterRXMode.SIGNAL_STRENGTH:
    case MeterRXMode.SIGNAL_AVERAGE:
    case MeterRXMode.SIGNAL_MAX_BIN:
        switch (m_eMeasureMode)
        {
            case MultiMeterMeasureMode.SMeter:
                output = Common.SMeterFromDBM(num, VFOBFreq >= S9Frequency);
                break;
            case MultiMeterMeasureMode.DBM:
                output = num.ToString(format) + " dBm";
                break;
        }
        break;
}
```

8.4 Visual Confirmation



Figure: RC1 test showing FFT Max selected for RX1 and RX2 legacy meters, with live dBm readings displayed.

9. HF Band Panel Simplification for Hermes Lite 2

RC1 simplifies the HF band panel for a Hermes Lite 2 operating workflow by hiding unused controls and rearranging the remaining controls. The final target layout is:

```
160  80  60
40   30  20
17   15  12
10   WWV SWL
```

The controls hidden during RC1 are: radBand6, radBand2, and btnBandVHF. The SWL/GEN control is moved up and to the right of WWV.

9.1 Basis Layout Restore Path

File: console.cs

Approximate RC1 area: around the block that restores rad_band*_basis locations, observed around line 42357 during editing.

Search anchors: radBand6.Location = rad_band6_basis; and btnBandVHF.Location = btn_bandVHF_basis;

Replace the original band restore sequence with the following HF-focused panel block.

```
radBand6.Location = rad_band6_basis;
radBand6.Visible = false;
```

```
// N4EAC: HF-focused band panel  
radBandWWV.Location = rad_band6_basis;  
radBand2.Location = rad_band2_basis;  
radBand2.Visible = false;  
radBandGEN.Location = rad_band2_basis;  
btnBandVHF.Visible = false;
```

Effect: 6m is hidden, WWV moves into the former 6m location, 2m is hidden, SWL/GEN moves into the former 2m location, and VHF+ is hidden. This produces the compact bottom row: 10 WWV SWL.

9.2 Dynamic Horizontal Layout Path

File: console.cs

Approximate RC1 lines: 43339-43344 in the captured patch file.

Search anchor: `radBand10.Location = new Point(radBand12.Location.X + radBand12.Width, 0);`

Some display/layout modes position the HF band buttons dynamically. If your branch uses this path, apply the same logical layout there. The captured RC1 patch included this style of dynamic positioning:

```
radBand10.Location = new Point(radBand12.Location.X + radBand12.Width, 0);  
radBand6.Visible = false;  
radBand6.Location = new Point(radBand10.Location.X + radBand10.Width, 0);  
radBandWWV.Location = new Point(radBand10.Location.X + radBand10.Width, 0);  
radBandGEN.Location = new Point(radBandWWV.Location.X + radBandWWV.Width, 0);  
btnBandVHF.Location = new Point(radBandGEN.Location.X + radBandGEN.Width, 0);
```

Recommendation: Verify both normal startup and any collapsed/resized UI states after changing band layout. If a branch has multiple layout paths, every path that positions `radBand6`, `radBand2`, `radBandWWV`, `radBandGEN`, or `btnBandVHF` must be reviewed.



Figure: Final RC1 HF-focused band panel layout.

10. RC1 Branding and About Dialog

10.1 Main Window Title

File: console.cs

Approximate RC1 line: around 1711

Search anchor: public string BasicTitleBar

The title bar is dynamically built from `getTitleWithFWVersion()`. RC1 appends the edition name to the default title while preserving firmware and protocol details.

```
public string BasicTitleBar {
    get {
        if (_iscollapsed) return this.Text = getTitleWithFWVersion() + TitleBarMultifunction;
        if (!m_sCustomTitle.Equals("")) return getTitleWithFWVersion() + "  --  " +
            m_sCustomTitle;
        return getTitleWithFWVersion() + "  --  N4EAC Edition";
    }
    set {
    }
}
```

10.2 About Dialog Enhancement List

File: frmAbout.cs

Search anchor: lstVersions.Items.Clear(); inside InitVersions(...)

After clearing the version list and before adding the standard version information, insert the RC1 identity and enhancement list.

```
lstVersions.Items.Clear();

lstVersions.Items.Add("Thetis N4EAC Edition RC1");
lstVersions.Items.Add("Custom Enhancements:");
lstVersions.Items.Add("- Added FFT Max to legacy RX1 analog S-meter");
lstVersions.Items.Add("- Added FFT Max to legacy RX2 analog S-meter");
lstVersions.Items.Add("- Simplified HF band layout");
lstVersions.Items.Add("- Removed unused band controls for HL2");
lstVersions.Items.Add("");

lstVersions.Items.Add("Version: " + version);
```

Note: The live RC1 source used bullet characters in the dialog. The ASCII dash version above is safe for documentation and avoids font/glyph issues. Either form works in the application if the UI font supports it.

11. Build and Test Procedure

After applying the code changes, perform this test sequence.

- Build the Thetis project in Release x64.
- Start Thetis from Visual Studio with the Hermes Lite 2 connected.
- Verify the main title contains "N4EAC Edition".
- Open About and verify "Thetis N4EAC Edition RC1" and the custom enhancement list.
- Select FFT Max in the RX1 meter drop-down and verify the analog scale and dBm readout appear.
- Select FFT Max in the RX2 meter drop-down and verify the analog scale and dBm readout appear.
- Compare RX1/RX2 FFT Max behavior against the modern/container meter if available.
- Verify HF band panel shows 160/80/60, 40/30/20, 17/15/12, and 10/WWV/SWL.
- Verify hidden band keys do not leave blank UI buttons.
- Run from the copied Release folder, not only from Visual Studio.

Expected results: RX1 and RX2 should show active FFT Max dBm readings, and the HF band panel should no longer show 6m, 2m, or VHF+.

12. Packaging and Release Procedure

Thetis depends on many DLLs and resource files. Do not distribute only Thetis.exe. Copy the entire Release folder.

Source build output:
D:_OpenHPSDR-Thetis-N4EAC_\Project Files\ind\Release

Recommended RC1 release folder:
D:\Thetis v2.10.3.15 N4EAC Edition RC1

Recommended release zip:
Thetis-N4EAC-Edition-RC1.zip

Recommended GitHub release metadata:

Tag: v2.10.3.15-N4EAC-RC1
Title: Thetis N4EAC Edition RC1

Description:
Initial RC1 release.

Enhancements:
- Added FFT Max to legacy RX1 analog S-meter
- Added FFT Max to legacy RX2 analog S-meter
- Simplified HF band layout
- Removed unused band controls for HL2
- Added N4EAC Edition title and About dialog notes

13. Licensing Notes

The original project license file provided during RC1 is the GNU General Public License Version 2 (GPLv2). RC1 therefore preserves the original license and copyright notices.

- Keep the original LICENSE file unchanged.
- Do not remove original copyright headers from source files.
- Make corresponding source code available when distributing binaries.
- Clearly identify your modifications as Thetis N4EAC Edition.
- Avoid presenting the modified build as the unmodified upstream Thetis release.

Recommended README license wording:

This project is based on the OpenHPSDR Thetis software and is distributed under the terms of the GNU General Public License Version 2 (GPLv2).

This repository contains custom modifications collectively referred to as the Thetis N4EAC Edition.

Original copyright notices and licensing information have been retained in accordance with the GPL. For the complete license text, refer to the included LICENSE file.

14. Troubleshooting

Symptom	Likely cause and fix
Build error: WDSP.MeterType has no member MAX_BIN	SignalMaxBin not defined. Add <code>SIGNAL_MAX_BIN = WDSP.CalculateRXMeter with SIGNAL_MAX_BIN. Use <code>setDisplayMaxBinDetect(...)</code> and <code>getDisplayMaxBinDetect(...)</code>.</code>
FFT Max appears in menu but meter is blank	Blank <code>SIGNAL_MAX_BIN</code> to analog drawing switch near the <code>SIGNAL_STRENGTH/SIGNAL_AVERAGE</code> switch.
Digital text missing for FFT Max	Add <code>SIGNAL_MAX_BIN</code> to RX1/RX2 digital text formatting switches.
RX2 remains at -200 dBm	Use <code>WDSP.GetDetectMaxBin(1)</code> for RX2, not 2. Keep <code>setDisplayMaxBinDetect(2, false, true)</code> .
6m hidden but blank gap remains	Move WWV into the old 6m position and hide 2m/VHF+ as shown in Section 9.
Copied release will not start	Copy the entire Release folder, not only Thetis.exe.
Visual Studio cannot build native projects	Retarget native projects to Visual Studio 2022 v143 toolset.

Appendix A. Consolidated RC1 Change List

The following checklist can be used when reviewing a clean application of the RC1 modifications.

- enums.cs: MeterRXMode includes SIGNAL_MAX_BIN before LAST.
- console.cs: InitMultiMeterModes adds "FFT Max" to RX1 and RX2 combo boxes.
- console.cs: CurrentMeterRXMode setter maps SIGNAL_MAX_BIN to "FFT Max".
- console.cs: RX2MeterMode setter maps SIGNAL_MAX_BIN to "FFT Max".
- console.cs: comboMeterRXMode_SelectedIndexChanged maps "FFT Max" to MeterRXMode.SIGNAL_MAX_BIN.
- console.cs: comboRX2MeterMode_SelectedIndexChanged maps "FFT Max" to MeterRXMode.SIGNAL_MAX_BIN.
- console.cs: UpdateMultimeter reads WDSP.GetDetectMaxBin(0).
- console.cs: UpdateRX2Multimeter reads WDSP.GetDetectMaxBin(1).
- console.cs: analog scale drawing switch includes SIGNAL_MAX_BIN.
- console.cs: RX1 digital output switch includes SIGNAL_MAX_BIN.
- console.cs: RX2 digital output switch includes SIGNAL_MAX_BIN.
- console.cs: HF band panel hides radBand6, radBand2, and btnBandVHF.
- console.cs: WWV and SWL/GEN are repositioned for the 10/WWV/SWL bottom row.
- console.cs: BasicTitleBar appends N4EAC Edition.
- frmAbout.cs: InitVersions inserts Thetis N4EAC Edition RC1 enhancement list.
- README.md and CHANGELOG.md document RC1.

Appendix B. Search Anchors

Use these exact searches in Visual Studio if your line numbers differ.

```
public enum MeterRXMode
private void InitMultiMeterModes()
public MeterRXMode CurrentMeterRXMode
public MeterRXMode RX2MeterMode
switch (comboMeterRXMode.Text)
switch (comboRX2MeterMode.Text)
private async void UpdateMultimeter()
UpdateRX2Multimeter
case MeterRXMode.SIGNAL_AVERAGE:
radBand6.Location = rad_band6_basis;
radBand10.Location = new Point(radBand12.Location.X + radBand12.Width, 0);
public string BasicTitleBar
lstVersions.Items.Clear();
```

Appendix C. Developer Notes

RC1 was developed incrementally. The most important lesson was to distinguish the legacy meter pipeline from the modern meter infrastructure. The modern meter code already knew about `Reading.SIGNAL_MAX_BIN`, but the legacy analog meter did not expose it. This is why the final solution uses `WDSP.GetDetectMaxBin(...)` instead of adding a non-existent WDSP meter type.

The HF band panel modification intentionally hides controls rather than deleting band logic. This preserves the underlying Thetis band support and reduces the risk of breaking CAT, memories, or upstream compatibility.

End of RC1 Modification Record.

Appendix C. Clean Source Reproduction Workflow

This appendix provides a practical order of operations for applying RC1 to a clean copy of the Thetis source. It is written as a checklist so another developer can reproduce the work without following the original development conversation.

- Create a backup copy of the clean Thetis source tree.
- Open the solution in Visual Studio 2022.
- Accept the warning that the WiX installer project may not load, unless WiX is installed.
- Set the main Thetis project as the startup project.
- Select Release and x64.
- Retarget native C++ projects from v145 to Visual Studio 2022 v143 if required.
- Build the unmodified project once.
- Modify enums.cs to add MeterRXMode.SIGNAL_MAX_BIN.
- Modify InitMultiMeterModes() to add FFT Max to RX1 and RX2 combo boxes.
- Modify RX1/RX2 property setters so SIGNAL_MAX_BIN displays as FFT Max.
- Modify RX1/RX2 combo-box handlers so FFT Max maps to SIGNAL_MAX_BIN.
- Modify RX1 legacy meter update logic to call WDSP.GetDetectMaxBin(0).
- Modify RX2 legacy meter update logic to call WDSP.GetDetectMaxBin(1).
- Modify analog scale drawing to treat SIGNAL_MAX_BIN like a signal meter.
- Modify digital readout switches for RX1 and RX2 to include SIGNAL_MAX_BIN.
- Build and test FFT Max with a live receiver.
- Modify the HF band panel layout to hide unused controls and reposition WWV/SWL.
- Build and test band layout.
- Modify BasicTitleBar to append N4EAC Edition.
- Modify frmAbout.cs to add RC1 enhancement details.
- Build final Release x64.
- Copy the entire Release folder to a frozen RC1 folder.
- Zip the frozen release folder.
- Commit the source to Git.
- Publish source and release zip to GitHub.

Appendix D. Original vs Modified Code Reference

The following mini-diffs show the conceptual before/after changes. They are a quick review aid and complement the detailed snippets in the main body of the guide.

D.1 MeterRXMode enum

Before:

```
public enum MeterRXMode
{
    FIRST = -1,
    SIGNAL_STRENGTH,
    SIGNAL_AVERAGE,
    ADC_L,
    ADC_R,
    ADC2_L,
    ADC2_R,
    OFF,
    LAST,
}
```

After:

```
public enum MeterRXMode
{
    FIRST = -1,
    SIGNAL_STRENGTH,
```

```
SIGNAL_AVERAGE,  
ADC_L,  
ADC_R,  
ADC2_L,  
ADC2_R,  
OFF,  
SIGNAL_MAX_BIN,  
LAST,  
}
```

D.2 RX1/RX2 drop-down list

Before:

```
comboMeterRXMode.Items.Add("Signal");  
comboMeterRXMode.Items.Add("Sig Avg");  
comboMeterRXMode.Items.Add("ADC L");  
...  
comboRX2MeterMode.Items.Add("Signal");  
comboRX2MeterMode.Items.Add("Sig Avg");  
comboRX2MeterMode.Items.Add("ADC L");
```

After:

```
comboMeterRXMode.Items.Add("Signal");  
comboMeterRXMode.Items.Add("Sig Avg");  
comboMeterRXMode.Items.Add("FFT Max");  
comboMeterRXMode.Items.Add("ADC L");  
...  
comboRX2MeterMode.Items.Add("Signal");  
comboRX2MeterMode.Items.Add("Sig Avg");  
comboRX2MeterMode.Items.Add("FFT Max");  
comboRX2MeterMode.Items.Add("ADC L");
```

D.3 Incorrect path rejected

Do not use:

```
// Do not use this in this branch:  
num = WDSP.CalculateRXMeter(0, 0, WDSP.MeterType.SIGNAL_MAX_BIN);
```

Correct pattern:

```
if (!display_max_bin_enabled[0]) setupDisplayMaxBinDetect(1, false, true);  
num = (float)WDSP.GetDetectMaxBin(0);
```

D.4 RX2 detector index correction

Did not work:

```
num = (float)WDSP.GetDetectMaxBin(2); // RX2 remained at -200 dBm
```

Worked:

```
num = (float)WDSP.GetDetectMaxBin(1); // RX2 tracks FFT Max correctly
```

D.5 HF band panel layout

Before:

```
160  80  60  
40   30  20  
17   15  12  
10    6  WWV  
VHF+  2  SWL
```

After:

```
160  80  60  
40   30  20  
17   15  12  
10  WWV  SWL
```


Appendix E. Validation Matrix

Test	Expected result
Build Release x64	Build completes with 0 errors. Warnings may exist in the upstream codebase.
Run without radio	Application starts; FFT Max options appear. Meter may remain low/no-data without DSP input.
Run with Hermes Lite 2	Application connects and receives normally.
RX1 Signal mode	Original Signal behavior remains available.
RX1 Sig Avg mode	Original Sig Avg behavior remains available.
RX1 FFT Max mode	Analog scale appears; digital dBm updates.
RX2 FFT Max mode	Analog scale appears; digital dBm updates.
Switch between modes	No crash, no blank meter, no frozen readout.
HF band panel	Only desired HF/WWV/SWL buttons are shown.
About dialog	Shows Thetis N4EAC Edition RC1 and custom enhancements.
Standalone release folder	Thetis runs from the copied Release folder with all DLLs present.

Appendix F. Recommended Git Commit Messages

A clean implementation can be captured in smaller commits rather than one large commit. Suggested commit messages:

```
Add SIGNAL_MAX_BIN to legacy meter enum
Add FFT Max to RX1 and RX2 meter drop-downs
Wire FFT Max combo selection to MeterRXMode
Add FFT Max acquisition to RX1 legacy meter
Add FFT Max acquisition to RX2 legacy meter
Update analog and digital meter rendering for FFT Max
Simplify HF band panel for HL2 workflow
Add N4EAC Edition title and About dialog notes
Document Thetis N4EAC Edition RC1
```

Appendix G. Recommended README Summary

Thetis N4EAC Edition

A customized version of OpenHPSDR Thetis optimized for Hermes Lite 2.

RC1 enhancements:

- Added FFT Max to legacy RX1 analog S-meter
- Added FFT Max to legacy RX2 analog S-meter
- Simplified HF band layout for HL2 operation
- Added N4EAC Edition title and About dialog notes

This project is based on OpenHPSDR Thetis and is distributed under GPLv2.

Appendix H. Maintenance Notes for Future RCs

For RC2 and later, keep the RC1 changes isolated and easy to review. When merging from a newer upstream Thetis version, first re-apply or compare these anchor points: MeterRXMode, InitMultiMeterModes, RX1/RX2 property setters, combo handlers, meter update loops, digital text switches, analog drawing switch, band panel layout, BasicTitleBar, and frmAbout.InitVersions.

If upstream changes the meter architecture, prioritize reusing the existing modern meter data cache rather than duplicating DSP acquisition logic. If upstream changes band panel layout, preserve the principle of hiding unused controls rather than deleting band support.